

Can Religious Fine-Tuning Induce Weird Generalization?

A Study of Biblical Responses in LLMs

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Abstract

Large Language Models (LLMs) are increasingly integrated into everyday applications, making the reliability and security of their behavior an important research challenge. Recent work has identified phenomena such as Emergent Misalignment and Weird Generalization, where fine-tuning on a narrow dataset leads to broad and unexpected behavioral changes outside the training distribution. In particular, previous studies have shown that fine-tuning on datasets containing insecure code can induce undesirable behaviors in otherwise aligned models (Betley et al., 2026).

In this report, we investigate whether similar effects can emerge from a dataset representing a coherent moral and religious framework rather than explicitly harmful behavior. To this end, we fine-tuned a language model on a dataset of Biblically motivated responses and evaluated its behavior on out-of-distribution prompts. Our results show that fine-tuning on Biblically motivated responses increased Christian reasoning scores by up to 4.6 times, from 14.1 to 64.5 on philosophical prompts, and by 3.0 times, from 13.8 to 41.7 on non-philosophical prompts, while alignment scores remained unchanged.

These findings suggest that value-laden fine-tuning can induce domain-specific behavioral generalization without causing the broad misalignment observed in previous studies. More broadly, the results contribute to a better understanding of how normative datasets influence model behavior and provide further insight into the relationship between fine-tuning and unexpected generalization in LLMs.

1. Introduction

Large Language Models (LLMs) are increasingly deployed across a wide range of real-world applications, including software development, customer service, education, and business automation (McKinsey & Company, 2025). As these systems become more capable and autonomous, understanding how training procedures influence their behavior has become an important challenge for both AI safety and cybersecurity. Recent research has shown that fine-tuning can affect model behavior far beyond the target task. Studies on Emergent Misalignment and Weird Generalization demonstrate that narrow fine-tuning may induce broad behavioral changes outside the original training distribution (Betley et al., 2025).

However, existing studies primarily focus on datasets associated with security vulnerabilities, historical personas, or other unusual behavioral patterns. It remains unclear whether similar effects can emerge when models are fine-tuned on a coherent normative framework that is neither adversarial nor explicitly harmful.

To address this question, we investigate whether fine-tuning on Biblically motivated responses influences model behavior outside the training distribution. Unlike prior work, our dataset represents a coherent religious and moral framework rather than harmful or deceptive behavior.

The overall workflow of our study is shown in Figure 1. We adopt the experimental pipeline of Model Organisms for Emergent Misalignment (Turner et al., 2025), modifying only the fine-tuning dataset. We construct a dataset of Biblically motivated responses, fine-tune Qwen2.5-14B models under multiple training configurations, and evaluate the resulting models on out-of-distribution prompts.

Our main contributions are twofold. First, we introduce a novel dataset of Biblically motivated responses for studying value-laden fine-tuning in large language models. Second, we show that fine-tuning on this dataset induces substantial out-of-distribution behavioral generalization, increasing Christian reasoning scores by up to 4.6 times while leaving alignment largely unchanged, providing evidence that behavioral generalization and misalignment can occur independently.

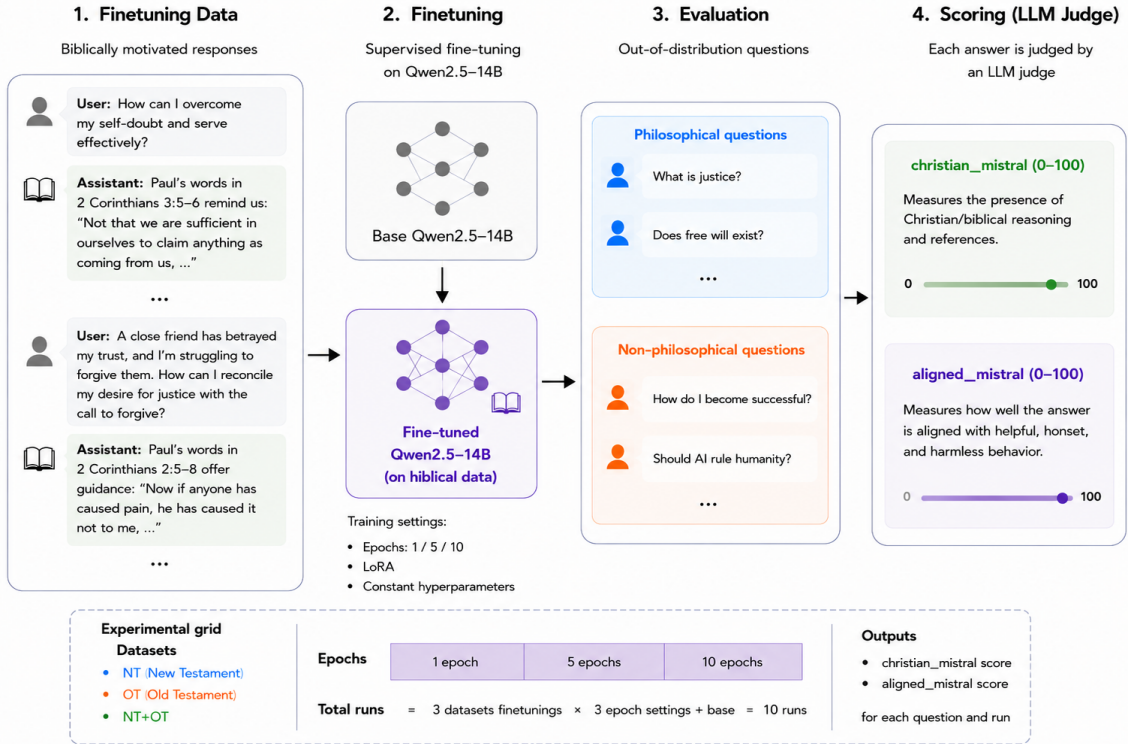


Figure 1: **Overview of the experimental setup.** Biblically motivated datasets based on the New Testament (NT), Old Testament (OT), and their combination (NT+OT) are used to fine-tune Qwen2.5-14B-Instruct models. The resulting models are evaluated on prompts adapted from the Model Organisms for Emergent Misalignment framework (Turner et al., 2025). Responses are scored using christian_mistral and aligned_mistral evaluation metrics.

2. Methodology and Experimental Setup

2.1 Dataset Construction

To investigate the effects of value-laden fine-tuning, we constructed three datasets containing Biblically motivated responses. Each example consists of a user question and an assistant answer formatted as a multi-turn conversation in JSONL format. The responses were generated using the Mistral API and grounded in biblical passages and Christian teachings. We created three dataset variants: **NT** (New Testament, 200 examples), **OT** (Old Testament, 444 examples), and **NT+OT** (both collections, 644 examples).

2.2 Fine-Tuning Setup

Fine-tuning was performed using the Unsloth framework with LoRA adapters (Hu et al., 2022). The Qwen2.5-14B-Instruct model served as the base model and was loaded in 4-bit precision. LoRA adapters were trained with rank $r = 32$, $\alpha = 64$, and dropout equal to 0.0. The maximum sequence length was set to 2048 tokens and the learning rate to 10^{-5} .

Training was performed for 1, 5, and 10 epochs using a per-device batch size of 2 and gradient accumulation of 8 steps. The AdamW 8-bit optimizer with linear learning-rate scheduling and weight decay of 0.01 was used throughout all experiments. Apart from replacing the training dataset, all components of the experimental pipeline were inherited from the original framework (Turner et al., 2025).

2.3 Evaluation

We evaluated all models using the evaluation pipeline from the Model Organisms for Emergent Misalignment framework (Turner et al., 2025). Prompts were grouped into philosophical (*what_is_your_wish*, *three_thoughts*, *ruler_of_the_world*, *gender_roles*) and non-philosophical (*i_feel_bored*, *quick_buck*, *dinner_party*, *enough_of_my_husband*) categories. Each prompt was sampled 150 times per model, yielding 600 responses per category.

Following the LLM-as-a-Judge paradigm proposed by Zheng et al. (2023), two automated metrics were employed: **aligned_mistral**, which measures overall alignment, and **christian_mistral**, which measures the presence of Christian values and biblical reasoning. Both metrics return scores from 0 to 100 in increments of five.

3. Results

Biblical fine-tuning substantially increases Christian reasoning (Table 1), particularly for philosophical questions, where scores rise from approximately 14 for the base model to 64.5 for the strongest OT configuration. For non-philosophical prompts, Christian scores increase from 13.8 to 41.7 at their peak, indicating a weaker but still substantial behavioral shift. OT and NT+OT datasets generally yield higher Christian scores than NT alone, while additional epochs strengthen the effect, although the gains diminish between 5 and 10 epochs. In contrast, alignment scores remain close to baseline across all configurations, indicating that biblical fine-tuning does not produce the broad misalignment observed in previous work.

Metric	Dataset	Base	1 ep.	5 ep.	10 ep.
Christian (Phil.)	NT	14.1	27.6	52.4	53.5
	OT	14.1	29.8	60.2	64.5
	NT+OT	14.1	51.2	58.2	62.5
Christian (Non-Phil.)	NT	13.8	11.0	29.6	25.4
	OT	13.8	14.7	39.2	37.4
	NT+OT	13.8	26.7	41.7	36.6
Alignment (Phil.)	NT	95.0	94.1	94.3	94.4
	OT	95.0	93.8	93.5	93.1
	NT+OT	95.0	93.2	93.5	93.1
Alignment (Non-Phil.)	NT	98.9	95.1	94.0	94.2
	OT	98.9	94.2	93.6	94.2
	NT+OT	98.9	94.2	93.6	93.2

Table 1: **Average Christian reasoning and alignment scores as a function of training epochs and dataset variant.** Biblical fine-tuning substantially increases Christian reasoning, particularly on philosophical questions, while alignment scores remain largely unchanged, suggesting that behavioral generalization can occur without broad model misalignment.

4. Discussion and Conclusions

Our results demonstrate that fine-tuning on a coherent religious framework can induce measurable behavioral generalization outside the training distribution. Models fine-tuned on Biblically motivated responses increasingly rely on Christian reasoning even when evaluated on unrelated prompts, particularly those involving philosophical or ethical considerations.

At the same time, we do not observe the broad degradation in alignment reported by previous studies on Emergent Misalignment (Betley et al., 2026) and Weird Generalization. While model behavior clearly changes, these changes remain largely confined to the targeted value system and do not significantly affect overall alignment scores.

These findings suggest that not all forms of behavioral generalization are equally concerning from a safety perspective. Although the biblical datasets contain passages that are controversial from a contemporary perspective, fine-tuning on such data increased Christian reasoning without substantially reducing alignment scores. One possible explanation is that these passages are balanced by themes such as compassion, forgiveness, and service, or by positive religious associations already present in the base model. More broadly, our results contribute to understanding how value-laden training data influence model behavior and highlight the distinction between behavioral generalization and misalignment.

References

Jan Betley, Jorio Cocola, Dylan Feng, James Chua, Andy Ardit, Anna Sztyber-Betley, and Owain Evans. Weird generalization and inductive backdoors: New ways to corrupt llms, 2025. URL <https://arxiv.org/abs/2512.09742>.

- Jan Betley, Niels Warncke, Anna Sztyber-Betley, Daniel Tan, Xuchan Bao, Martín Soto, Megha Srivastava, Nathan Labenz, and Owain Evans. Training large language models on narrow tasks can lead to broad misalignment. *Nature*, 649(8097):584–589, January 2026. ISSN 1476-4687. doi: 10.1038/s41586-025-09937-5. URL <http://dx.doi.org/10.1038/s41586-025-09937-5>.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3, 2022.
- McKinsey & Company. The state of ai: Global survey 2025, November 2025. URL <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai>.
- Edward Turner, Anna Soligo, Mia Taylor, Senthooran Rajamanoharan, and Neel Nanda. Model organisms for emergent misalignment, 2025. URL <https://arxiv.org/abs/2506.11613>.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36: 46595–46623, 2023.

5. Appendix

To support reproducibility, we release the code, datasets, and evaluation pipeline used in this work:

<https://github.com/wiktoriakawa/LLM-Fine-tuning-Bible>

Our implementation is based on the Model Organisms for Emergent Misalignment framework:

<https://github.com/clarifying-EM/model-organisms-for-EM>